The effect of Corn cobs Xylo-Oligosaccharides on the microbiota composition and metabolites

By Gili Nachmani

Advisors: Dr. Faiga Magzal, Dr. Loai Basheer, Dr. Paula Pitashny

Tel-Hai College

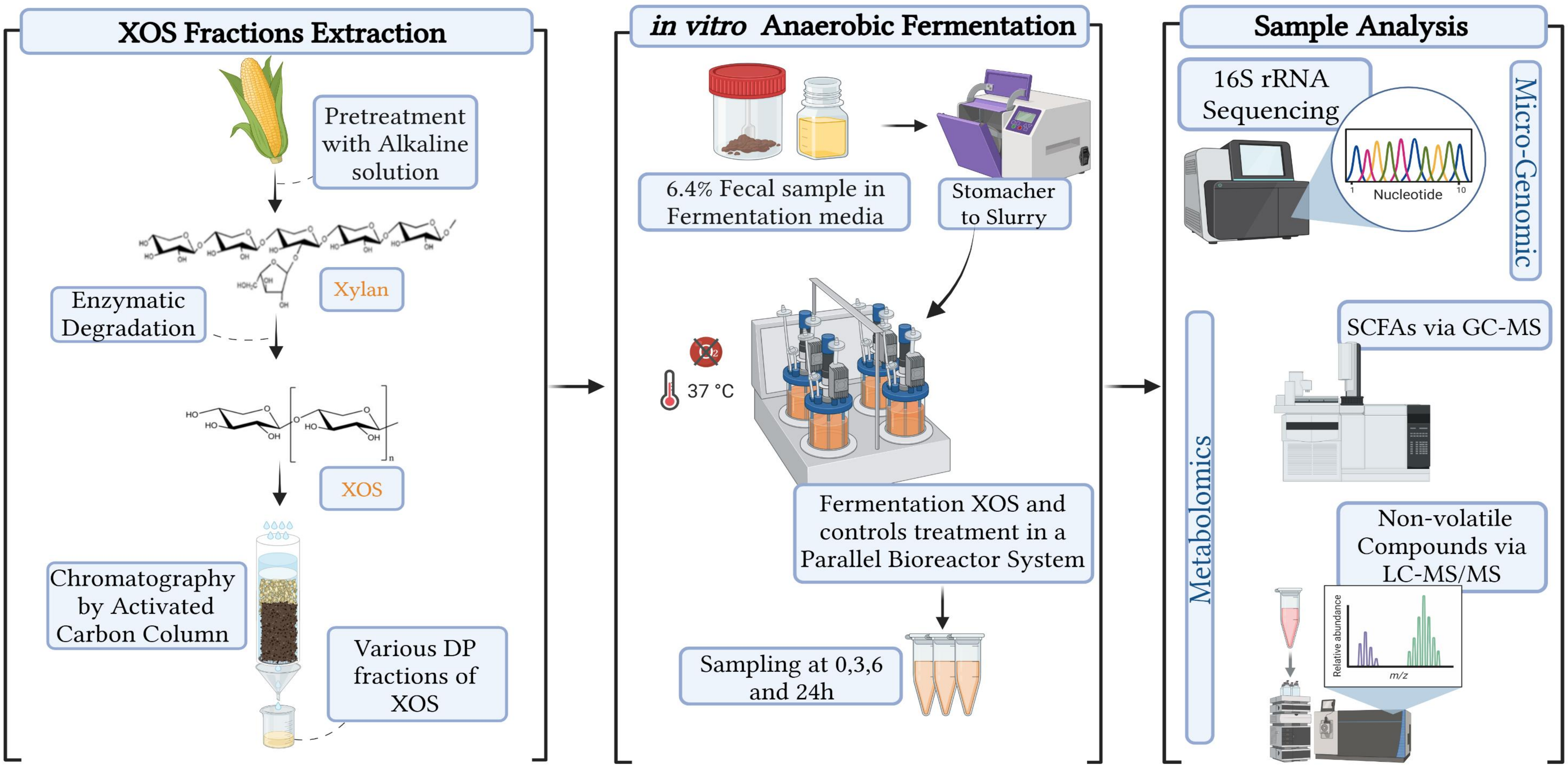
MIGAL Galilee Research Institute

Abstract

Background: The gut microbiome is essential for digestion, immunity, and overall health, and its composition is strongly influenced by diet[1]. Xylo-oligosaccharides (XOS), fibers derived from corn cobs, have emerged as promising prebiotics that beneficially shape microbial communities[2]. Besides, ongoing research reveals beneficial techno-functional properties of XOS in food[3]. Understanding how different XOS structures influence microbial activity could lead to targeted strategies for promoting gut health through their implementation in innovative food products.

Research Goals: To produce XOS from corn cobs, fractionate them by Degree of Polymerization (DP), and compare the prebiotic potential of high DP and short DP of XOS using an anaerobic *in vitro* system that simulates colonic conditions.

Methods



Results

Results of preliminary experiment: The effect of *in vitro* fermentation of commercial XOS, commercial FOS (positive control), and a fiber-free medium (negative control) on short-chain fatty acids (SCFAs) produced by gut microbiota.

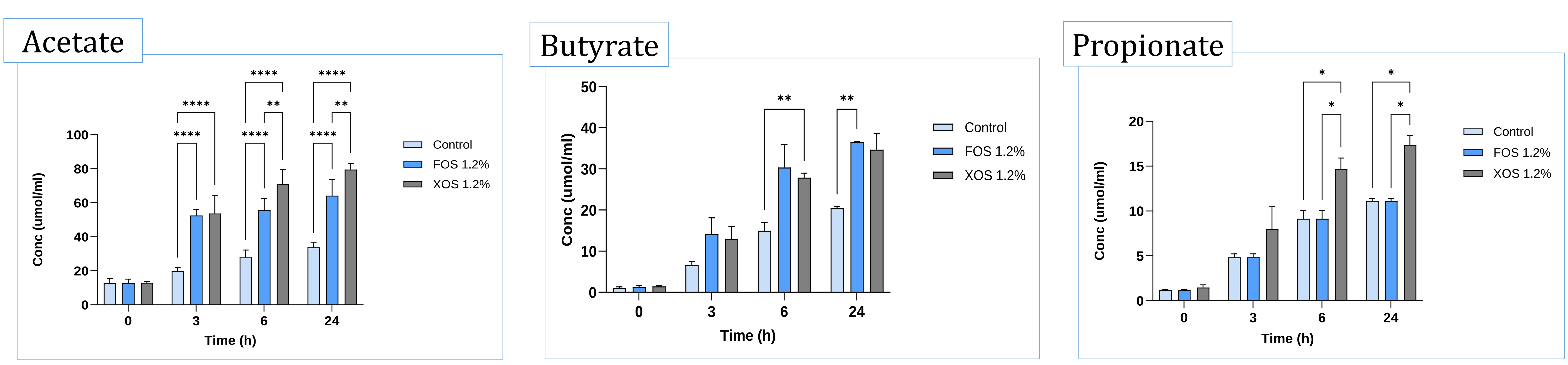


Figure 1: SCFAs concentrations of a control (light blue), 1.2% Commercial FOS (blue), and 1.2% Commercial XOS (gray) after 0,3,6 and 24 hours of fermentation. Data represent mean concentrations ± SD (n=3) and statistical analysis was performed using one-way ANOVA followed by Tukey's post hoc test (*=p<0.05,**=p<0.01,***=p<0.001,****=p<0.0001).

Conclusion

XOS extracted from corn cobs shows strong potential as prebiotics by **enhancing the production of SCFAs**- beneficial microbial metabolites. Comparing high DP and short DP of XOS will further clarify how polymerization degree shapes microbiome responses and supports gut health.

References

[1] Vemuri et al., *Microorganisms,* 2020.

[2] Valladares-Diestra et al., *Foods,* 2023.

[3] Shlomi Dean, M.Sc. Thesis, 2024.